

REVIEW

Targeting cellular senescence with senotherapeutics: senolytics and senomorphics

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The concept of geroscience is that since ageing is the greatest risk factor for many diseases and conditions, targeting the ageing process itself will have the greatest impact on human health. Of the hallmarks of ageing, cellular senescence has emerged as a druggable therapeutic target for extending healthspan in model organisms. Cellular senescence is a cell state of irreversible proliferative arrest driven by different types of stress, including oncogene-induced stress. Many senescent cells (SnCs) develop a senescent-associated secretory phenotype (SASP) comprising pro-inflammatory cytokines, chemokines, proteases, bioactive lipids, inhibitory molecules, extracellular vesicles, metabolites, lipids and other factors, able to promote chronic inflammation and tissue dysfunction. SnCs up-regulate senescent cell anti-apoptotic pathways (SCAPs) that prevent them from dying despite the accumulation of damage to DNA and other organelles. These SCAPs and other pathways altered in SnCs represent therapeutic targets for the development of senotherapeutic drugs that induce selective cell death of SnCs, specifically termed senolytics or suppress markers of senescence, in particular the SASP, termed senomorphics. Here, we review the current state of the development of senolytics and senomorphics for the treatment of age-related diseases and disorders and extension of healthy longevity. In addition, the challenges of documenting senolytic and senomorphic activity in pre-clinical models and the current state of the clinical application of the different senotherapeutics will be discussed.

Introduction

Due to improvements over the last century in sanitation, vaccines and health care, people are living longer than ever before. The World Health Organization predicts that the population of people over 60 years old will double before 2050 [1]. However, advancing age is recognized as one of the greatest risk factors associated with the development of chronic diseases [2,3]. In fact, over 90% of those above the age of 65 have at

least one chronic disease, while over 70% have at least two, highlighting the need for therapeutic interventions to prevent increased societal burden [4,5].

Ageing is driven by the progressive decline in the ability to manage various stressors. This leads to genomic instability, mitochondrial dysfunction, metabolic alterations, stem cell exhaustion and other deleterious attributes that contribute to homeostatic imbalance in

Abbreviations

ATM, ataxia telangiectasia mutated; CRC, colorectal carcinoma; HSP90, heat shock protein 90; IAP, inhibitor of apoptosis proteins; PPAR α , peroxisome proliferator-activated receptor alpha; PROTACs, proteolysis targeting chimeras; SASP, senescence-associated secretory phenotype; SCAP, senescent cell anti-apoptotic pathway; SnCs, senescent cells.

various tissues, driving age-associated pathologies and increasing susceptibility to death [2]. Cellular senescence is induced by different types of stress with age and contributes to age-associated pathologies, including diabetes, cardiovascular and Alzheimer's diseases, osteoarthritis, cancer and more [6–13]. As discussed in detail next, the demonstration that cellular senescence plays a causal role in multiple age-associated ailments highlights the potential of targeting senescent cells as a therapeutic strategy for the prevention and mitigation of many age-associated chronic diseases while extending healthspan.

Cellular senescence is the state of irreversible proliferative arrest triggered by persistent activation of stress-signalling mechanisms as a safeguard for malignant transformation. The paradox of cellular senescence as a protective mechanism and a driver of chronic disease can be attributed to unique metabolic alterations and their pathological, pro-inflammatory senescence-associated secretory phenotype (SASP). The SASP comprised pro-inflammatory cytokines, chemokines, proteases, bioactive lipids, inhibitory molecules and other factors, promoting a state of chronic inflammation and tissue dysfunction which drives age-related diseases (Fig. 1) [3,14–19]. Proteases present in the SASP promote degradation of the extracellular matrix, while pro-inflammatory molecules act in a paracrine manner, negatively affecting neighbouring cells by inducing cell death, the transition into a senescent state, development of SASP and promoting evasion of immune cell clearance [20–25]. SASP factors can also act in an autocrine manner that maintains a growth-arrested state while increasing SASP secretion [20,25,26]. While all senescent cells (SnCs) exhibit alterations in cell morphology and cycle arrest, many of them differentially regulate the exit from the cell cycle, activation of stress responses, pro-survival and anti-apoptotic pathways and exhibit variation in the production and profile of their SASP. Therefore, SnCs are incredibly diverse, exhibiting variation in metabolic and secretory profiles with some exhibiting no SASP secretion at all. This remarkable variation highlights the need to better characterize the different types of SnCs and the mechanisms that drive senescence and SASP secretion.

Our laboratory and others have demonstrated the therapeutic benefit of utilizing small molecules that specifically target SnCs, termed senotherapeutics, for the improvement of age-related diseases [27–30]. Proof of senotherapeutic efficacy for the mitigation of age-related diseases has been supported by research utilizing mouse models of progeria and chronic disease and their rapid transition from discovery to clinical

trials [6,27,31–33]. Senotherapeutics come in two classes: senolytics, which selectively kill SnCs or induce senolysis, and senomorphics, which attenuate the pathological SASPs to cause senostasis (Fig. 1). The majority of the senolytics identified to date promote the apoptosis of SnCs by targeting critical enzymes involved in pro-survival and anti-apoptotic mechanisms, such as p53, p21, Bcl-2 family proteins, Akt, PI3K, FOXO4 and others (Table 1) [31,34–39]. However, due to the diverse nature of SnCs, many of the identified senolytics exhibit specificity towards specific types or sub-populations of SnCs. Despite this heterogeneity, senolytics have demonstrated the ability to alleviate many chronic ailments such as atherosclerosis and other cardiovascular dysfunctions, liver fibrosis, stem cell function and others, along with a concomitant increase in healthspan [11,30,40–44].

In general, senomorphics suppress senescence by suppressing SASP expression via targeting NF- κ B, mTOR, IL-1 α , p38 MAPK and other signalling pathways (Table 2) [45–52]. In this manner, senomorphics, though less popular than senolytics, may be a preferred method of intervention as SnCs are also implicated in processes such as wound healing and development [53,54]. Consistent with the heterogeneity of SnCs, studies investigating the mechanistic action of senotherapeutics have observed that certain compounds have the dual capacity to function as both a senomorphic and a senolytic, in a cell-specific and even concentration-dependent manner [31]. Here, we will review currently reported senotherapeutic molecules speculating on mechanistic actions and the dual capabilities of some of these molecules, as well as their biological impacts.

Senolytics

Dasatinib and quercetin

The first senolytics reported were a drug combination of dasatinib (D) and quercetin (Q), discovered from a hypothesis-driven, bioinformatics-based study. Transcriptome analysis of senescent versus non-senescent human preadipocytes revealed that SnCs rely on senescent cell anti-apoptotic pathways (SCAPs) to protect them from apoptosis, including ephrin receptors, BCL-2/BCL-X_L family members, PI3K/AKT, HIF-1 α and others [36]. Dasatinib, an FDA-approved anti-cancer drug, is a tyrosine kinase inhibitor able to inhibit cell proliferation and migration and induce apoptosis. Quercetin is a naturally occurring flavonoid with diverse biological activities including interacting with a PI3K isoform and Bcl-2 family members. The

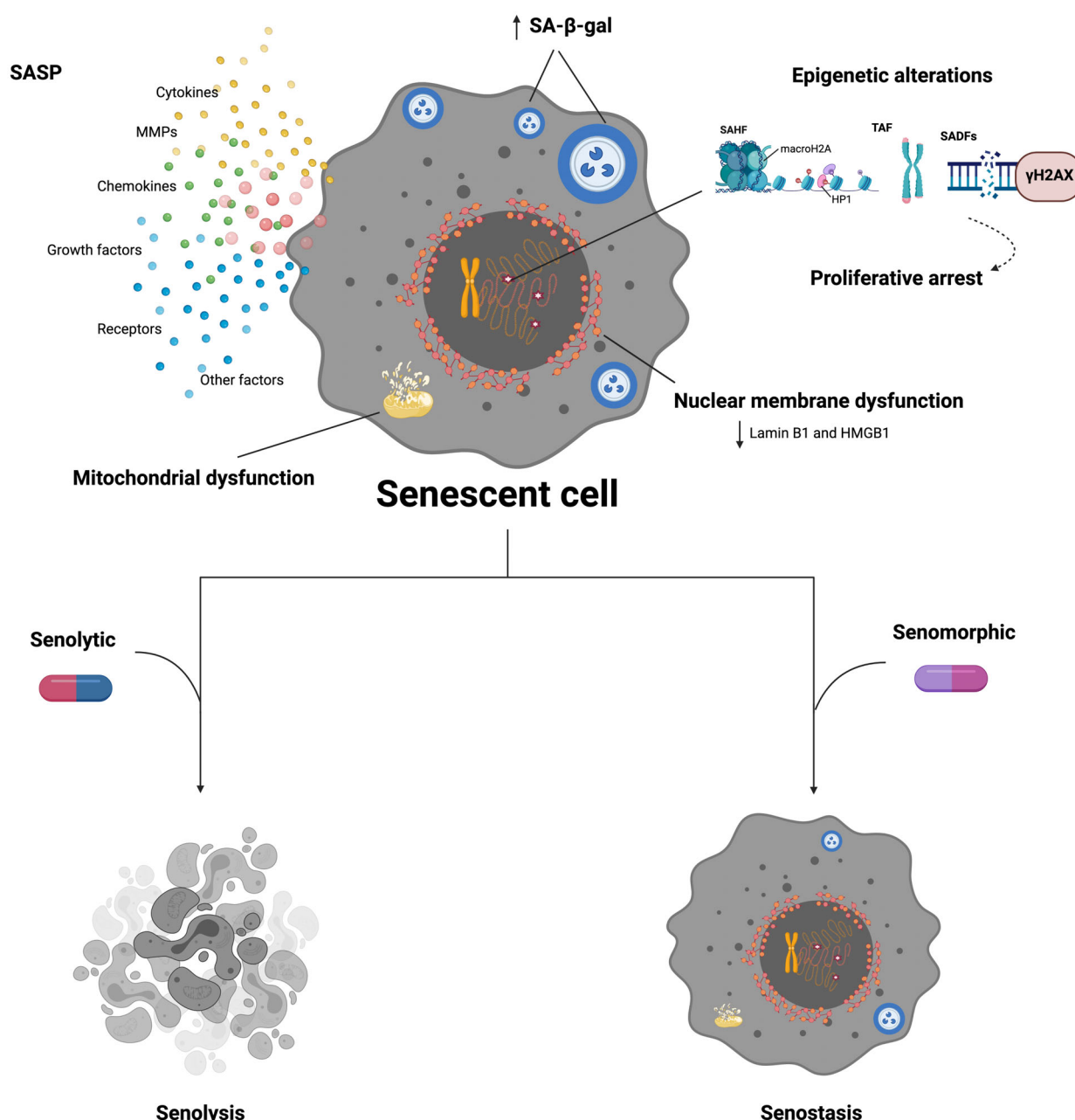


Fig. 1. General hallmarks of cellular senescence and action mechanisms of senolytic versus senomorphic compounds. Cellular senescence is accompanied by a variety of biochemical alterations that drive senescence-associated characteristics such as enlarged and irregular morphology; an increase in β -galactosidase activity termed the senescence-associated β -galactosidase activity (SA- β -gal); mitochondrial dysfunction, depletion of Lamin B1 resulting in nuclear membrane dysfunction; alterations in the epigenetic landscape; production of the senescence-associated phenotype and proliferative arrest. Metabolic alterations associated with the senescent state can be targeted using senotherapeutics. There are two kinds of senotherapeutics: senolytics, which induces senolysis in senescent cells, and senomorphics, which attenuate their pathological pro-inflammatory secretory phenotype to cause senostasis. MMPs, matrix metalloproteinases; SASP, senescence-associated secretory phenotype; SAHFs, senescence-associated heterochromatin foci; SADFs, senescence-associated DNA damage foci; TAFs, telomere-associated foci. Created with [BioRender.com](https://www.biorender.com).

D + Q combination induces apoptosis more efficiently by targeting more SCAPs than either drug alone in many types of SnCs [36]. D + Q treatment improves

many age-associated diseases and disorders in mice including frailty, osteoporosis, hepatic steatosis, insulin resistance, neurodegeneration, vasomotor

Table 1. List of current senolytics and their mechanisms of action.

Class	Compound	Mechanism of action
BCL-2 family inhibitors	Dasatinib (D) + Quercetin (Q)	SCAPs
	ABT-737	Inhibit BCL-2, BCL-XL, BCL-W
	ABT-263 (Navitoclax)	Inhibit BCL-2, BCL-XL, BCL-W
	A-1331852	Inhibit BCL-XL
	A-1155463	Inhibit BCL-XL
HSP90 inhibitors	17-DMAG (alvespimycin)	Disrupt HSP90-AKT interaction
	Geldanamycin	Inhibit HSP90
	17-AAG (tanespimycin)	Inhibit HSP90
	Ganetespib	Inhibit HSP90
Targeting p53	FOXO4-DRI	Disrupt FOXO4-p53 interaction
	UBX0101	Disrupt MDM2/p53 interaction
	RG7112 (RO5045337)	Disrupt MDM2/p53 interaction
	P5091	USP7 inhibitor
	P22077	USP7 inhibitor
Natural products and their analogues	Fisetin	BCL-2, PI3K/AKT, p53, NF- κ B and more
	Curcumin	Down-regulate Nrf2 and NF- κ B pathways
	o-Vanillin (curcumin metabolite)	Down-regulate Nrf2 and NF- κ B pathways
	EF-24 (curcumin analogue)	BCL-2 family
	Piperlongumine and its analogues (compounds 47–49)	Unclear, OXR1, NF- κ B
	GL-V9	Unclear, alkalize lysosome and elevate ROS levels
Cardiac glycosides	Proscillaridin A	Inhibit Na ⁺ /K ⁺ -ATPase
	Ouabain	Inhibit Na ⁺ /K ⁺ -ATPase, up-regulate BCL2-family protein NOXA
	Ouabagenin	Inhibit Na ⁺ /K ⁺ -ATPase
	Digoxin	Inhibit Na ⁺ /K ⁺ -ATPase, up-regulate BCL2-family protein NOXA
	Bufalin	Inhibit Na ⁺ /K ⁺ -ATPase
Galactose modified pro-drugs	K-Strophanthin	Inhibit Na ⁺ /K ⁺ -ATPase
	Strophanthidin	Inhibit Na ⁺ /K ⁺ -ATPase
	SSK1	SA- β -galactosidase
	Pro-drug A (JHB75B)	SA- β -galactosidase
	Nav-Gal	SA- β -galactosidase
PROTACs	5FURGa	SA- β -galactosidase
	PZ15227	Degrade BCL-XL
	ARV825	Degrade BRD4
Miscellaneous	Fenofibrate	PPAR α agonist

Table 1. (Continued).

Class	Compound	Mechanism of action
	Azithromycin	Induce autophagy and glycolysis
	Roxithromycin	NOX4
	Tamatinib (R406)	Syk inhibitor, FAK and p38MAPK
	MitoTam	Reduce mitochondrial membrane potential, inhibit OXPHOS
	Panobinostat	Histone deacetylase inhibitor
	AT-406	Inhibitor of c-IAP1, c-IAP2 and XIAP

dysfunction, pulmonary fibrosis, chronic kidney disease and skeletal muscle dysfunction [55]. To date, the senolytic combination of D + Q has been tested in several human clinical trials for the treatment of idiopathic pulmonary fibrosis (NCT02874989), chronic kidney disease (NCT02848131), skeletal health (NCT04313634), hematopoietic stem cell transplant survivors (NCT02652052) and Alzheimer's disease (NCT04063124).

Bcl-2 family member inhibitors

Certain members of the BCL-2 family of proteins, including BCL-2, BCL-X_L and BCL-W, are key regulators of cell survival that confer senescent cell resistance to apoptosis. The BCL-2 family members are up-regulated during senescence, while their down-regulation or inhibition in SnCs induces apoptotic cell death. Many BCL-2 inhibitors have been identified as novel senolytics, including ABT-737, ABT-263 (Navitoclax), A-1331852 and A-1155463. ABT-737, an inhibitor of BCL-W, BCL-X_L and BCL-2, can trigger senolysis in senescent MEF and IMR90 fibroblast cells [56]. In mice, ABT-737 treatment eliminated SnCs in the lung and epidermis [56]. However, ABT-737 is not orally bioavailable and has low aqueous solubility; therefore, structural optimization was performed and its derivative, ABT-263, was developed as an orally bioavailable pan-BCL inhibitor [57]. The senolytic effect of ABT-263 depends on the cell type. It selectively induced apoptotic cell death of senescent HUVECs, IMR-90 and MEF cells, but not primary human preadipocytes [34]. When administered to sublethally irradiated or normally aged mice, ABT-263 rejuvenated aged hematopoietic stem cells, in part,

Table 2. List of current senomorphics and their mechanisms of action.

Class	Compound	Mechanism of action
NF- κ B inhibitors	Rapamycin	Unclear; mTOR, Nrf2, NF- κ B
	Metformin	Unclear; NF- κ B, Nrf2/GPx7, DICER1, MBNL1, STAT3, AMPK, SIRT1, insulin/IGF-1, mTOR, etc.
	Resveratrol and STACs	SIRT1
	Aspirin	Unclear
	SR12343	Inhibition of IKK/NF- κ B activation
	SB203580	p38MAPK inhibitor
	UR13756	p38MAPK inhibitor
	BIRB796	p38MAPK inhibitor
	MK2.III	MK2 kinase inhibitor
	PF-3644022	MK2 kinase inhibitor
JAK/STAT inhibitors	Ruxolitinib	JAK1/2 inhibitor
ATM inhibitors	KU-55933	ATM inhibitor
	KU-60019	ATM inhibitor
Statins	Atorvastatin	HMG-CoA reductase inhibitor
	Pravastatin	HMG-CoA reductase inhibitor
	Pitavastatin	HMG-CoA reductase inhibitor
	Simvastatin	HMG-CoA reductase inhibitor, inhibition of protein prenylation
Natural products	Apigenin	IRAK1/I κ B α /NF- κ B
	Kaempferol	IRAK1/I κ B α /NF- κ B
	Quercetin	Proteasome activator
	Quercetin caprylate	Proteasome activator
	Epigallocatechin gallate (EGCG)	Suppress p53 acetylation
	Genistein	mTOR, AMPK
	Oleuropein	NF- κ B
	aglycone	
	Hydroxytyrosol	NF- κ B

through its effective clearance of SnCs [32]. In cancer therapy, ABT-263 reduces the number of senescent cancer cells caused by the chemotherapeutic agent doxorubicin, possibly reducing the incidence of cancer relapse and metastasis by attenuating the detrimental SASPs [58]. However, ABT-263 has severe adverse effects such as thrombocytopenia and neutropenia, which likely result from its pan-BCL inhibition of BCL-2, BCL-X_L and BCL-W [59]. More specific BCL-2 inhibitors are speculated to have reduced side effects by minimizing the toxicity on platelets and neutrophils [60]. BCL-X_L inhibitors A-1331852 and A-1155463 were also senolytic in senescent HUVECs and human lung fibroblasts [61]. Nonetheless, it remains to be investigated whether they have fewer side effects than ABT-263. Although these BCL-2 inhibitors are the first generation of senolytics, the side effects on

platelets will likely limit their further translational use in clinic.

HSP90 inhibitors

Heat shock protein 90 (HSP90) is a family of abundantly expressed chaperone proteins that play a key role in the stabilization and degradation of hundreds of client proteins, many of which are involved in cell proliferation, angiogenesis and oncogenesis [62]. Therefore, inhibiting HSP90 has been an effective approach for cancer chemotherapy [63]. Recently, several HSP90 inhibitors, including 17-DMAG (alvespimycin), geldanamycin and 17-AAG (tanespimycin), were found to display senolytic activities in oxidative stress-induced senescent *Ercc1*^{-/-} MEFs, human lung fibroblasts (IMR90 and WI-38) and vascular endothelial cells (HUVECs) [30]. Treatment of *Ercc1*^{-/-}Δ progeroid mice with 17-DMAG reduced tissue senescence, delayed the onset of multiple age-related symptoms and extended healthspan. Mechanistically, 17-DMAG promoted apoptosis of SnCs, in part, by disrupting the HSP90-AKT interaction to destabilize the active form of AKT [30].

p53 pathway targeting compounds

Transcription factor p53 plays a critical role in many biological processes, including DNA repair, cellular stress response, cell growth, apoptosis and the initiation and maintenance of senescence [64]. Recently, p53 and its regulatory networks have become an alternative target for senolytic development.

FOXO4-p53 inhibitors

The forkhead box protein O4 (FOXO4) transcription factor is highly expressed in SnCs and prevents cell death by binding to and sequestering nuclear protein p53 [65]. On this basis, a senolytic FOXO4-D-retro-inverso (FOXO4-DRI) peptide was identified that disrupts this FOXO4-p53 interaction, allowing p53 to translocate to the cytosol, inducing cell-intrinsic apoptosis [37]. FOXO4-DRI selectively kills senescent Leydig cells and alleviates age-related hypogonadism in aged male mice. Administration of FOXO4-DRI in the *Xpd*^{TTD/TTD} mouse model of accelerated ageing or naturally aged mice also alleviated ageing symptoms such as skin phenotypes, fur density and renal function [66]. In addition, FOXO4-DRI prevented chemotoxicity in mice, evidenced by its inhibition of doxorubicin-induced senescence and the associated weight loss and liver damage [66].

MDM2/p53 inhibitors

E3 ligase murine double minute 2 (MDM2) is a major negative regulator of p53 that targets p53 for proteasomal degradation [67]. UBX0101 is an inhibitor of MDM2/p53 interaction able to induce apoptosis of senescent chondrocytes in an osteoarthritis mouse model [39]. However, the Phase II clinical study of UBX0101 to treat osteoarthritis did not result in statistical significance compared to the placebo [68]. RG7112 (RO5045337) is another small molecule inhibitor of MDM2/p53 interaction used to restore p53 activity. RG7112 can selectively kill senescent intervertebral disc cells and decrease SASP factors, including IFN- γ , IL-6 and CCL24, through senolysis in culture [69]. When tested in intact human discs *ex vivo*, RG7112 reduced SnCs in intact human discs and improved disc matrix homeostasis [69].

USP7 inhibitors

Ubiquitin-specific peptidase 7 (USP7) deubiquitinates MDM2 and prevents its degradation by the ubiquitin–proteasome system, thereby providing another approach to stabilize and up-regulate p53 [70]. In senescent WI-38, IMR-90 fibroblasts, renal epithelial cells and HUVECs, p53 expression is significantly lower than basal level. However, USP7 inhibitors P5091 and P22077 were able to selectively kill SnCs by reducing MDM2 expression and up-regulating p53, leading to the induction of pro-apoptotic proteins PUMA, NOXA and FAS as well as the inhibition of BCL-X_L/BAK. In mice, P5091 can also eliminate doxorubicin-induced SnCs and suppress SASP factors [71].

It is important, however, to note that p53 plays a dual role in promoting and sometimes inhibiting senescence, depending on cell line and levels and types of cell stressors [72,73]. For example, in the context of replicative or mild genotoxic stress, p53 up-regulation is pivotal for the induction of senescence. Conversely, some SnCs down-regulate p53, making them resistant to apoptosis. Clearly, more studies are needed to dissect the complex molecular mechanisms of p53 on regulating senescence.

Natural products and their analogues

Fisetin

Fisetin is a naturally occurring flavonoid found in many fruits, vegetables, flowers and teas. Diverse pharmacological effects of fisetin have been reported, including

antioxidant, anti-inflammatory, anti-cancer, anti-diabetic, antibacterial, antiviral and neuroprotective activities [74,75]. Fisetin has diverse mechanisms of action on multiple molecular targets and signalling pathways including BCL-2, PI3K/AKT, p53, NF- κ B and more [76]. Recently, fisetin was shown to possess senolytic activity. Fisetin can selectively trigger apoptotic cell death in radiation-induced senescent HUVECs, oxidative-stress induced senescent MEFs and genotoxin-induced senescent human fibroblasts. In addition, like the combination of D + Q, fisetin can reduce the extent of senescence in human tissue explants [61,77]. Intermittent treatment of the *Ercc1*^{−/Δ} mouse model of accelerated ageing with fisetin reduced cellular senescence in multiple tissues. Also, in wild-type old mice, fisetin treatment improved tissue homeostasis, alleviated age-related pathologies and extended median and maximum lifespan [77]. Currently, fisetin has entered several clinical trials for chronic kidney disease (NCT03325322), skeletal health (NCT04313634), osteoarthritis (NCT04210986), COVID-19 (NCT04476953, NCT04537299, NCT04771611), survivors of childhood cancers (NCT04733534) and frailty (NCT03675724).

Curcumin

Another natural product curcumin also showed senolytic activity against senescent human intervertebral disc cells by down-regulating the Nrf2 and NF- κ B pathways [78]. EF-24, an analogue of curcumin with improved bioavailability, has more potent senolytic activity in a variety of SnCs including human fibroblasts, endothelial cells, renal epithelial cells and preadipocytes. The senolytic effect of EF-24 appears to be through increasing degradation of the anti-apoptotic BCL-2 family proteins [79].

Piperlongumine

The natural product piperlongumine isolated from genus *Piper* was identified as another class of senolytics in senescent WI38 fibroblasts induced by ionizing radiation, replicative exhaustion or oncogenic *Ras* expression [80]. Piperlongumine selectively induces the death of SnCs by directly binding to oxidation resistance 1 (OXR1), leading to its proteasomal degradation and subsequent increased ROS production [81]. Piperlongumine also inhibits lung tumour growth via inhibition of the NF- κ B signalling pathway [82]. However, the precise senolytic mechanisms of action of senolytic piperlongumine are still unclear. Through a series of structural modifications, several piperlongumine

analogues (compounds 47–49) were developed with improved senolytic activity [83].

GL-V9

GL-V9, a synthetic flavonoid analogue of wogonin, has been demonstrated to have pro-apoptotic, anti-invasive, anti-metastatic, anti-tumour and autophagic effects in various cancer cells [84–87]. Recently, GL-V9 was shown to be a potential senolytic that preferentially killed senescent breast cancer cells and replication-induced senescent MEFs in a ROS-dependent manner [88]. In senescent MDA-MB-231 cells, GL-V9 at 10 μ M was able to alkalinize the lysosome and elevate ROS levels, aggravating mitophagy deficiency, to induce apoptosis. When injected intravenously in MMTV-PyMT mice, GL-V9 decreased cellular senescence in epirubicin-induced tumours. However, no senolytic effects were observed with wogonin, the parent compound of GL-V9. Structurally, a tertiary amine group is a distinguishing component of GL-V9, which may facilitate its selective localization in acidic lysosomes, thus contributing to the ROS elevation and apoptotic killing in SnCs [89]. In another study, a non-lethal dosage of GL-V9 (2 μ M) was found to promote cellular senescence in malignant T-cell lines, including Jurkat, HuT-102 and HuT-78 [90]. More mechanistic studies are needed to identify the targets of GL-V9 and understand its induction of senescence and cell death through mitotic catastrophe at varying doses.

Cardiac glycosides

Recently, two independent studies identified a family of cardiac glycosides as senolytics by high-throughput screening, including proscillaridin A, ouabain, ouabagenin, digoxin, bufalin, k-stropanthin and strophanthidin [91,92]. Ouabain is a broad-spectrum senolytic in a variety of SnCs undergoing replicative, chemotherapy or oncogene-induced senescence [92]. Intermittent ouabain treatment *in vivo* reduced the burden of SnCs accumulated in irradiated mice and diminished the level of local inflammation and immune infiltration when administered to old mice [92]. Similarly, digoxin can preferentially kill senescent cancer cells induced to senescence by bleomycin or palbociclib [91]. Administration of digoxin with senescence-inducing chemotherapy drugs, gemcitabine or doxorubicin provided synergistic effects at inhibiting tumour growth and inducing complete regression of tumours in mice, demonstrating the potential benefit of using this senolytic as a complementary anti-cancer therapy

[91]. These cardiac glycosides may exert their senolytic activities by inhibiting the Na^+/K^+ -ATPase, causing destabilization of the SnC membrane potential and intracellular acidification. Both ouabain and digoxin also can activate the expression of genes of the pro-apoptotic BCL2 family, most notably NOXA [92].

Galactose-modified senolytic pro-drugs

One common characteristic that SnCs share is elevated activity of the lysosomal β -galactosidase (SA- β -gal). Therefore, pro-drugs with a cleavable galactose moiety attached to a cytotoxic compound can be processed specifically in SnCs by the increased SA- β -gal activity, leading to the release of the cytotoxic compound and selective killing of senescent cells. Several senolytic pro-drugs have been developed with this approach.

Gemcitabine, a clinically used chemotherapeutic, was modified with an acetyl galactose moiety and a non-toxic aromatic spacer to yield pro-drug SSK1 [93]. Notably, SSK1 selectively kills senescent mouse and human cells, irrespective of senescence inducers and cell types, highlighting the advantage of the pro-drug strategy over classic senolytics. In aged mice, SSK1 also effectively cleared SnCs in different tissues, decreased SASP expression and improved physical function [93].

A galactose-modified duocarmycin pro-drug (pro-drug A, JHB75B) was developed by linking two galactose groups to a cytotoxic seco-duocarmycin analogue dimer [94]. Pro-drug A preferentially killed SnCs with high levels of SA- β -gal activity. It also displayed senolytic activity towards a wide spectrum of SnCs induced by etoposide or doxorubicin treatment, irradiation or serial passage. In mice, pro-drug A also eliminated bystander senescent cells arising from anti-cancer therapies and preneoplastic SnCs [94].

This pro-drug approach also has been applied to reduce the significant haematological toxicity of navitoclax. A senolytic navitoclax pro-drug (Nav-Gal) was generated by conjugating the BCL2 family inhibitor navitoclax with an acetylated galactose group [95]. Nav-Gal showed effective senolytic activity in chemotherapy-induced senescent A549 lung cancer cells and SK-Mel-103 melanoma cells with a higher senolytic index than navitoclax, which suggests that the pro-drug is preferentially activated in SnCs. Concomitant treatment with Nav-Gal and chemotherapeutic cisplatin significantly inhibited tumour growth in both subcutaneous tumour xenografts and orthotopic mouse models of chemotherapy-induced senescence in the context of lung carcinoma. Remarkably, compared to navitoclax, Nav-Gal treatment did not cause

thrombocytopenia and showed reduced platelet toxicity in mice, providing evidence that galactose-conjugated pro-drugs are an effective strategy to reduce navitoclax-driven thrombocytopenia [95].

The chemotherapeutic agent 5-fluorouridine (5-FU) exerts its cytotoxicity by interfering with DNA replication via inhibition of thymidylate synthase. This anti-cancer drug was modified with a β -D-galactosyl moiety to afford an inactive pro-drug 5-fluorouridine-5'-O- β -D-galactopyranoside (5FURGal) [96]. 5FURGal is converted into active cytotoxic 5-fluorouracil in senescent human IMR90 fibroblasts, but not in proliferating or quiescent cells, which leads to its selective killing of SnCs. When tested in mice, 5FURGal reduced senescence and improved frailty, muscle tissue and stem cell functions, cognitive function and overall mouse survival with minimal toxicity and high tolerability [96].

PROTACs

Proteolysis targeting chimeras (PROTACs) are bifunctional molecules comprising a ligand capable of binding to a target protein and a ligand capable of recruiting an E3 ubiquitin ligase coupled by an optimal linker. PROTACs can attach to and bring the target protein and the E3 ligase into proximity, resulting in the ubiquitination and subsequent destruction of the target protein via the ubiquitin proteasome system. In contrast to traditional inhibitors, PROTAC chemicals degrade target proteins in a catalytic manner rather than inhibiting their activity by a competitive- or occupancy-driven process. Because of their catalytic action, PROTACs require less drug exposure, resulting in longer activity and lower toxicity when compared to traditional inhibitors. The PROTAC technology has garnered attention in recent years and has been successfully applied in drug discovery to target different types of proteins related to various diseases [97,98].

The PROTAC approach has been applied to increase the efficacy and specificity of senolytics. For example, a novel PROTAC-based senolytic PZ15227 was designed by tethering ABT263 to a pomalidomide moiety with an empirically optimized linker. PZ15227 binds to BCL-X_L, causing it to be degraded by the cereblon (CRBN) E3 ligase. In aged mice, PZ15227 was more effective than ABT-263 at clearing SnCs, but with less toxicity to platelets in which the cereblon E3 ligase is minimally expressed [37]. ARV825 comprised the small molecule BRD4, binding moiety OTX015, and E3 ligase binder pomalidomide. It was originally designed as a degrader of BRD4 to suppress cell proliferation and induce apoptosis in cancer cells [99]. In a high-throughput senolytic screening,

ARV825 was able to robustly induce senolysis in SnCs [100]. The senolytic activity of ARV825 may result from the PROTAC-based degradation of BRD4 and subsequent exacerbation of DNA double-strand breaks, eventually leading to autophagy-induced apoptotic cell death [100]. In obesity-induced hepatocellular carcinoma mouse models, ARV825 treatment eliminated senescent hepatic stellate cells in the liver and reduced the development of liver cancer. In addition, in immunocompromised mice, ARV825 was able to eliminate chemotherapy-induced SnCs and increase the efficacy of chemotherapy against xenograft tumours [100]. It is expected that more PROTAC senolytics will be identified in the future. However, due to the large molecular weights of PROTACs, additional pharmacokinetic and pharmacodynamic studies are necessary to ensure their adequate absorption when delivered *in vivo*.

Miscellaneous

Fenofibrate is an agonist of peroxisome proliferator-activated receptor alpha (PPAR α), which regulates lipid metabolism. In a cell-based screening using human chondrocytes, fenofibrate was found as a novel senolytic and pro-autophagy agent. In studies examining human osteoarthritis and ageing primary chondrocytes, fenofibrate treatment up-regulated PPAR α expression, reduced chondrocyte senescence, increased autophagic flux and induced selective elimination of SnCs [28].

Azithromycin and roxithromycin, two FDA-approved macrolide antibiotics used to treat various bacterial infections, were identified as a new class of senolytic in a senolytic screening of clinically approved drugs. Interestingly, azithromycin also demonstrated a biphasic effect on mitochondrial oxygen consumption rate, with an inhibitory activity at lower concentration (50 μ M), while stimulatory activity was observed at a higher concentration (100 μ M) [101]. Metabolic analysis found that azithromycin can induce autophagy and glycolysis, possibly attributing to the death of SnCs. Roxithromycin also was shown to attenuate pulmonary fibrosis in bleomycin-induced mice by targeting senescent cells mediated by the NOX4 pathway [102].

The Syk inhibitor tamatinib (R406) was identified as a novel senolytic from a high-throughput screening using a replicative senescence model of human dermal fibroblasts. Tamatinib was shown to target multiple regulatory pathways for senescent cell survival, including the focal adhesion kinase (FAK) and p38 mitogen-activated protein kinase (MAPK) pathways [103].

SnCs can also be targeted through interfering with their mitochondrial integrity and ATP homeostasis.

MitoTam, a mitochondria-targeted tamoxifen, was reported as a new senolytic that selectively induced senescent cell death through decreasing mitochondrial membrane potential and inhibiting mitochondrial oxidative phosphorylation (OXPHOS), consequently exacerbating mitochondrial integrity [104].

As described earlier, numerous studies have shown that chemotherapeutic agents induce senescence not only in normal cells, but also in certain cancer cells. The accumulation of therapy-induced senescent cancer cells and their persistent SASP secretion can negatively influence the tumour microenvironment and promote aggressiveness and stemness of cancer cells, leading to drug resistance, tumour relapse and metastasis [105]. For example, cisplatin or taxol treatment led to senescence in non-small cell lung cancer and head and neck squamous cell carcinoma cell lines [106]. Panobinostat, an FDA-approved histone deacetylase inhibitor, then was used as a post-chemotherapeutic agent following cisplatin or taxol treatment. Interestingly, panobinostat caused cell death more efficaciously than re-treating cancer cells with successive cycles of chemotherapy. The senolytic activity of panobinostat was attributed to its ability to reverse the expression patterns observed in senescence, with decreased expression of BCL-X_L and increased acetylated H3 [106].

Inhibitor of apoptosis proteins (IAP) are a group of anti-apoptotic regulators of caspases and cell death. In senescent mesenchymal stem cells, the anti-apoptotic proteins c-IAP1, c-IAP2 and XIAP are up-regulated and more resistant to pro-apoptotic stimuli than non-senescent cells [107]. AT-406, an inhibitor of these IAPs, specifically induced the apoptosis of SnCs and attenuated the progression of osteoarthritis in rats [107].

Senomorphics

Senomorphic treatment provides an alternative pharmacological approach to target cellular senescence. Senomorphics suppress the detrimental effects of SASP components secreted by senescent cells without causing cell death. The first senomorphics were mainly discovered by serendipity including rapamycin, metformin, resveratrol and aspirin.

Rapamycin

Rapamycin (also known as sirolimus) is a macrolide compound isolated from the bacterium *Streptomyces hygroscopicus* found on Easter Island and named after the indigenous name of the island, Rapa Nui [108]. Initially, rapamycin was used as an anti-fungal agent.

Later, its immunosuppressive [109] and anti-proliferative [110,111] properties were discovered, and it was approved by the FDA for the prevention of organ rejection with kidney transplantation, as well as for the treatment of lymphangioleiomyomatosis. Recently, additional biological activities of rapamycin were revealed, including reducing cellular senescence, suppressing SASPs and extending lifespan. Numerous studies have shown that rapamycin could reduce cellular senescence and suppress SASP markers in a variety of mouse, rat and human cell lines, induced to senesce using a variety of different approaches and agents [112]. Rapamycin has been recognized as one of the most well-established senomorphics that regulate senescence and SASPs. In several invertebrate models, rapamycin increased the lifespan of yeast, worms and flies [113]. More intriguingly, the NIA Interventions Testing Program (ITP) in 2009 found that administering rapamycin late in life extended both median and maximum lifespan of both male and female mice [114]. Similarly, many other *in vivo* studies have demonstrated that rapamycin can alleviate age-related dysfunctions, delay tumour onset, decrease rate of ageing and increase lifespan in different mouse models [115]. However, rapamycin has some side effects including metabolic dysregulation, thrombocytopenia, hyperlipidaemia and impaired wound healing [115].

The mechanistic studies of rapamycin led to the discovery of TOR gene and the mTOR (mechanistic or mammalian target of rapamycin) pathway [116–118]. The mTOR protein is the central component of two different multiprotein complexes: TORC1 and TORC2 [119]. Rapamycin inhibits TORC1 activity more specifically in association with the intracellular protein FKBP12. To date, the roles of the mTOR signalling pathway have expanded from coordinating cell growth, proliferation, survival and autophagy to regulating ageing, cellular senescence and lifespan [120–122]. The senomorphic and longevity effects of rapamycin are largely related to its inhibition of mTOR signalling by reducing phosphorylation of S6K and 4E-BP downstream of TORC1 [120–122]. Other potential or secondary mechanisms by which rapamycin may regulate senescence and lifespan are being explored, for example, activating the Nrf2 pathway [123] and decreasing NF- κ B activity to reduce IL-1 α production [124]. It is proposed that the aforementioned side effects of rapamycin may result from its off-target inhibition of mTORC2; therefore, development of novel rapamycin analogues or rapalogues may hold promise to overcome the side effects of rapamycin while improving its potency, solubility and pharmacokinetic properties [125]. It is also to be

determined whether other mTOR pathway inhibitors and rapalogues such as everolimus, temsirolimus, deforolimus, ridaforolimus and zotarolimus may exhibit senomorphic activities.

Metformin

Metformin is a synthetic biguanide first-line drug that was originally approved for the treatment of type 2 diabetes. To date, the therapeutic effects of metformin have expanded into age-related disorders, including insulin resistance, obesity, liver diseases, cardiovascular diseases, cancer, renal diseases and neurodegenerative diseases [126]. In addition, numerous data in diverse model organisms have proved that metformin is effective in suppressing cellular senescence and SASPs and attenuating multiple age-associated dysfunctions. For example, metformin treatment reduced cellular SA- β -gal activity and down-regulated the expression of senescence and SASP factors in different types of SnCs such as human diploid fibroblasts [46,127], lens epithelial cells [128] and mouse olfactory ensheathing cells [129]. Metformin also increases healthspan and lifespan in different model organisms, including *Caenorhabditis elegans* [130,131], crickets [132] and mice [133–135], independent of its effect on diabetes. Intriguingly, metformin treatment significantly reduced all-cause mortality and incidence of age-related diseases of diabetic patients compared with healthy individuals, suggesting it is geroprotective [136,137]. Metformin is the first drug to be tested for its age-targeting effects in the TAME (targeting ageing by metformin) clinical trial [138].

Despite its long history of clinical application and extensive mechanistic studies, the exact mechanisms of action of metformin remain elusive. Nonetheless, its versatile beneficial effects suggest that metformin is a pleiotropic senomorphic able to modulate multiple pathways and molecular targets in a complex manner. For example, metformin could suppress SASPs by inhibiting the phosphorylation of I κ B and IKK α/β to block the translocation of NF- κ B to the nucleus [46]. Low-dose metformin treatment reduced senescence through transcriptional up-regulation of Nrf2-mediated glutathione peroxidase 7 (GPx7) in human diploid fibroblasts and mesenchymal stem cells [127]. Metformin also decreases cellular senescence and inflammatory SASPs in a DICER1-dependent manner [139]. In renal tubular epithelial cells, metformin reduced senescence by up-regulating the expression of RNA-binding protein MBNL1 and miR-130a-3p and down-regulating the STAT3 pathway, thus alleviating diabetic nephropathy [140]. However, the senomorphic

effect of metformin may only partially account for its therapeutic potential in the prevention and treatment of various ageing-associated pathological conditions since metformin has been found to be able to influence all of the hallmarks of biological ageing and their interconnectivity directly or indirectly. These include mediating nutrient-signalling pathways through the activation of AMPK and SIRT1 and down-regulation of insulin/IGF-1 signalling and mTORC1, attenuating oxidative damage and genome instability by increasing DNA-damage like response and DNA repair, enhancing proteostasis by augmenting autophagy and inhibiting protein synthesis. Additionally, metformin treatment improved mitochondrial dysfunction via mitochondrial complex I inhibition and PGC-1 α up-regulation, inducing stem cell rejuvenation and delaying stem cell ageing, transcriptional and epigenetic regulation via histone and DNA modifications, reducing telomere shortening by activating via telomeric repeat-containing RNA, modulating the gut microbiota to improve metabolism and reduce inflammation and more [141,142].

Resveratrol and STACs

Silencing information regulator 2-related enzyme 1 (sirtuin1, SIRT1) is an NAD⁺-dependent deacetylase that regulates many signalling and transcriptional pathways involved in senescence and ageing. The expression of SIRT1 is diminished with age, whereas its increased expression is sufficient to reduce cellular senescence and extend lifespan in many organisms [143]. Despite its various health benefits, SIRT1 activator resveratrol has a biphasic effect on cellular senescence. At low concentrations (usually below 10 μ M), resveratrol is a senomorphic or antioxidant able to prevent cellular senescence and suppress SASPs. For example, resveratrol was shown to prevent cellular senescence of endothelial progenitor cells by activation of telomerase through the PI3K-Akt signalling pathway [144] and alleviate senescence in neural cells via activation of SIRT1-mediated STAT3 signalling [145]. It was also shown to inhibit cellular senescence of human bone marrow stromal stem cells [146] and suppress SASP factors by inhibiting NF- κ B and up-regulating Nrf2 pathways in vascular smooth muscle cells from aged rhesus monkeys [147]. However, at higher concentrations (typically over 25 μ M), resveratrol acts as a pro-oxidant triggering growth arrest and induction of senescence and/or apoptotic death in multiple cell lines [148]. Moreover, although resveratrol has been shown to extend lifespan in yeast, nematodes, fruit flies and short-lived fish, its effect in mice varies,

as it only extends lifespan of mice on high-fat diets and not those on standard diets [149,150]. Besides doses, other variables have also been proposed to account for the divergent or opposing effects of resveratrol on cellular senescence and other biological processes, including cell and tissue types, timing and routes of dosing, sex and genetic background and diet compositions.

Despite the poor bioavailability and stability of resveratrol, numerous other sirtuin-activating compounds (STACs) with distinct structures have been developed such as SRT1720, SRT17201460 and SRT17202183 (second generation), and STAC-5, STAC-9 and STAC-10 (third generation) [151]. Notably, the SIRT1 activator, SRT1720, is 1000-fold more potent than resveratrol and can extend the lifespan of mice on a high-fat or standard diet [152,153]. SRT2104 displays good tolerability and bioavailability in humans and has entered several clinical trials for treatment of age-related diseases.

Aspirin

Beyond its classic use as a non-steroidal anti-inflammatory drug, aspirin also has diverse therapeutic benefits in many age-related diseases such as atherosclerosis, arthritis, osteoporosis and cancer [154]. Intriguingly, aspirin is able to extend lifespan and healthspan in worms and mice [155]. Aspirin can also inhibit senescence and SASPs, suggesting a potential mechanism for its healthy ageing effects. For example, aspirin delayed the onset of replicative senescence of endothelial cells by increasing nitric oxide synthesis and decreasing oxidative stress, consequently up-regulating telomerase activity [156]. In a mouse model mimicking chemotherapy for childhood cancer, aspirin reduced the levels of senescence markers and ameliorated the long-term adverse effects in doxorubicin-treated juvenile mice [157]. Conversely, aspirin treatment increased the senescence of SW620 and HCT116 human colorectal carcinoma (CRC) cells. This senescence-inducing effect of aspirin occurred by causing an energy imbalance in CRC cells and subsequent activation of SIRT1 and AMPK activity [158]. These results suggest that the effects of aspirin on senescence may vary with cellular context, stress type and/or treatment doses.

NF- κ B inhibitors

As a master transcription factor, NF- κ B controls the expression of many genes including inflammatory cytokines and SASP factors. More importantly, NF-

κ B activation is not only associated with, but also drives cellular senescence and the ageing process [159,160]. Genetic inhibition of NF- κ B reduced tissue senescence and attenuated ageing phenotypes in different mouse models [161–163]. Pharmacological inhibition of NF- κ B with a peptide IKK inhibitor reduced cellular senescence *in vitro* and *in vivo*, and attenuated progeroid symptoms and pathologies of *Ercc1*^{−/Δ} mice [162]. To overcome the limitation of low stability and bioavailability commonly seen for peptides, we recently developed a small molecule IKK/NF- κ B inhibitor SR12343 through virtual drug screening followed by structural optimization [164]. Pharmacological inhibition of IKK/NF- κ B activation by SR12343 reduced senescence and SASP factors in senescent fibroblast cells induced by oxidative stress or the genotoxic agent etoposide. Moreover, *in vivo* treatment with SR12343 reduced senescence, improved tissue pathologies, and extended healthspan in naturally aged mice as well as in several models of accelerated ageing [165]. These results suggest that inhibition of the over-activated NF- κ B pathway associated with senescence and ageing is a promising senotherapeutic intervention to treat age-related diseases. Given that NF- κ B is closely coordinated with several collateral pathways involved in longevity and ageing by positive or negative mechanisms, such as insulin/IGF-1, mTOR, sirtuins, FoxOs, Nrf2 and DNA damage response [166], many senomorphics, especially natural products, exert their anti-senescence effects partially through direct or indirect NF- κ B inhibition.

p38MAPK inhibitors

The MAPK pathways are evolutionarily conserved pathways which deliver stress and damage signals to the nucleus to direct cell fate, including growth, proliferation, differentiation, survival and motility. The MAPKs comprise three subfamilies: extracellular signal-regulated kinases (ERKs), c-Jun N-terminal kinases (JNKs) and p38MAPKs. Among these, p38MAPKs are activated by a wide range of cellular stressors, including pro-inflammatory cytokines, ultraviolet irradiation, heat shock, oxidative stress and genotoxic stress, therefore, they are also referred to as stress-activated MAPK. Most of these stressors are capable of inducing senescence, thus, it is not surprising that the p38MAPK pathway also plays an important role in promoting and maintaining cellular senescence and SASPs [167]. Diverse senescence-inducing stimuli activate p38MAPK and constitutive p38MAPK activation is sufficient to trigger irreversible cell cycle arrest and induce cellular senescence and

SASP development [167]. Further, p38MAPK inhibitor SB203580 markedly reduced the secretion of most SASP factors and mitigated the paracrine effects of SnCs induced by irradiation or oncogenic RAS expression [168]. In mice exposed to a sub-lethal radiation, p38 was highly activated in bone marrow hematopoietic cells, and SB203580 treatment significantly reduced p16^{INK4a} expression and SA- β -gal activity, attenuating radiation-induced bone marrow suppression [169]. Likewise, more specific p38MAPK inhibitors UR13756 and BIRB796 were highly effective at blocking the SASP in SnCs [50]. Notably, inhibition of p38MAPK downstream target effector MK2 kinase by two small molecules MK2.III and PF-3644022 suppressed the SASP secretion of SnCs, suggesting SASP can be regulated by inhibiting a reduced number of pathways downstream p38 MAPK [50]. ERKs and JNKs also regulate cellular senescence and SASPs, but they serve dual or opposing roles under different circumstances.

JAK/STAT inhibitors

The Janus kinase/signal transducer and activator of transcription (JAK/STAT) signalling pathway also plays an important role in the regulation of inflammation and SASP expression. The JAK/STAT pathway is highly up-regulated in SnCs compared to non-senescent cells. Inhibition of the JAK pathway genetically by siRNA or pharmacologically by the JAK1/2 inhibitor ruxolitinib suppressed SASP production [18]. Ruxolitinib treatment in old mice reduced systemic inflammation, improved metabolic function and alleviated frailty [18]. Ruxolitinib was also found to inhibit progerin-induced senescence *in vitro* and reduced premature ageing phenotypes in Zmpste24-deficient mice [170]. The JAK2/STAT3 axis also participates in oestrogen-related senescence of bone marrow mesenchymal stem cells, and ruxolitinib treated, ovariectomized mice reduced expression of SASP factors, alleviated senescence and enhanced osteogenic differentiation [171].

ATM inhibitors

Ataxia telangiectasia mutated (ATM) is a key protein kinase triggered by genomic dysfunction that is involved in DNA damage. Persistent or constitutive ATM activation contributes to cellular senescence, whereas ATM depletion reduces the secretion of SASP factors [172]. We found that ATM and its downstream effectors are persistently elevated in *Ercc1*^{-/-} progeria and naturally aged mice, which is concomitant with hyperactive NF- κ B signalling. Genetic reduction of

ATM in *Ercc1*^{-/-} mice reduced NF- κ B activity, cellular senescence and attenuated age-related bone and intervertebral disc pathologies, leading to an extension of healthspan [173]. More importantly, inhibiting ATM pharmacologically with KU-55933 down-regulated NF- κ B activation and reduced senescence and SASP expression [173]. Consistently, inhibition of ATM activity by another compound KU-60019 alleviated cellular senescence by recovering mitochondrial function in normal fibroblasts [174] and accelerated ageing cells [175]. These results indicate that the use of ATM inhibitors can suppress genotoxic stress-induced senescence to ameliorate age-related diseases. However, given the importance of ATM in DNA repair response, the inactivation or long-term inhibition of ATM may increase cancer risk. Therefore, ATM activity should be finely tuned to achieve beneficial effects in reducing senescence and detrimental SASPs while minimizing the potential risk.

Statins

Statins are a class of HMG-CoA reductase inhibitors that are clinically used as cholesterol-lowering drugs for patients with hypercholesterolemia. Apart from cholesterol reduction, some statins, including atorvastatin, pravastatin and pitavastatin, were reported to inhibit oxidative stress-induced endothelial senescence. These statins increase phosphorylation of Akt at Ser473 resulting in the up-regulation of endothelial nitric oxide synthase, SIRT1, and catalase [176]. Another HMG-CoA reductase inhibitor simvastatin was also shown to decrease SASP expression in senescent human fibroblasts and suppress the cell non-autonomous cancer-promoting effect of SASPs in breast cancer cells. Simvastatin acts by inhibiting protein prenylation, and consequently, the activity of two Rho GTPase family members, Rac1 and Cdc42, that regulate actin dynamics and cell proliferation [177]. However, it should be noted that side effects of statins include muscle problems, increased risk of type 2 diabetes and liver damage.

Natural products

Many natural products, especially polyphenols, have well-documented antioxidant and anti-inflammatory activities; therefore, it is likely that some of them may have the potential to suppress senescence and inflammatory SASPs. Several naturally occurring flavonoids, particularly apigenin and kaempferol, were demonstrated to significantly inhibit SASP production in bleomycin-induced senescent BJ fibroblasts [178]. The

SASP inhibitory effect of these flavonoids was mediated by inhibiting NF- κ B p65 activity via the IRAK1/I κ B α signalling pathway and by blocking the expression of I κ B ζ . Oral administration of apigenin also effectively reduced elevated levels of SASP and I κ B ζ expression in the kidneys of aged rats [178]. The natural antioxidant quercetin and its derivative quercetin caprylate were identified as proteasome activators, able to reduce the percentage of β -galactosidase positive cells and maintain the young morphology of primary human fibroblasts [179]. Epigallocatechin gallate (EGCG), an abundant compound in green tea, prevents serial passage- and H₂O₂-induced senescence in human dermal fibroblasts via suppression of p53 acetylation [180]. Genistein, a primary isoflavone from soy products, induced autophagy and attenuated senescence in adriamycin-stimulated senescent human vascular cells by inhibiting mTOR signalling and activating AMPK pathway [181]. Oleuropein aglycone and hydroxytyrosol, two abundant phenols in olive fruits, are able to reduce SA- β -gal-positive cells and p16^{INK4a} protein expression, as well as attenuate markers of SASPs, including IL-6, COX-2 and metalloproteases, in cultured human fetal lung MRC5 fibroblasts and neonatal human dermal fibroblasts (NHDF) [182]. However, many natural products are notoriously known for their complex modes of action with multiple targets. Poor bioavailability is also a common concern for many polyphenol natural products.

Many other compounds have been reported to extend lifespan in different models; however, whether their anti-ageing effects are through senomorphic activity is still to be investigated. For example, the senomorphic activity of nordihydroguaiaretic acid (NDGA) was shown only at a single concentration in senescent *Ercc1*^{-/-} MEFs [30], though it extends the lifespan of mice, particularly in males tested in the NIA ITP [183]. Likewise, lifespan extension and other beneficial health effects of acarbose and 17- α -oestradiol have been reported, but whether it is related to its senomorphic activity remains unclear [183]. Two Ca²⁺ channel blockers, loperamide and nifedipine, and dopamine antagonist fluspirilene also displayed a senomorphic effect on primary MEFs; however, only a single concentration was tested [30].

Discussion

The identification of the first-ever senolytics, D + Q, and their rapid transition from discovery to demonstration of efficacy in mouse models to clinical trials has fuelled further senotherapeutic discovery. The majority of senotherapeutics reported to date are mechanistically

described as being senolytic and thus selectively kill SnCs by targeting critical molecular pathways associated with the senescence phenotype such as anti-apoptotic and pro-survival networks. In contrast, senomorphics, which suppress pathological SASP secretion without killing senescent cells, could be as or more effective and safer alternative to senolytics. However, identifying mechanistic action of senotherapies, senomorphic or senolytic is challenging, especially *in vivo*, and is often dependent on the SnC type and the concentration of the senotherapeutic utilized.

Senolytic compounds clearly have demonstrated efficacy in alleviating progeroid symptoms increasing healthspan and reducing incidences of disease in model organisms. However, only a few of the senotherapeutics discussed earlier, such as the combination of D + Q, UBX010 and fisetin, have advanced into clinical trials. Unlike UBX-010, which targets the p53/MDM2 interaction, fisetin and D + Q target multiple senescence-associated signalling pathways, thus, inducing apoptosis more effectively. In addition, these senolytics also could work through additional mechanisms that may directly or indirectly influence multiple hallmarks of ageing including senescence.

SnCs are heterogeneous in regard to their transcriptional, metabolic and SASP profiles between species, cell types and even at the individual cell basis, resulting in differential reliance on various senescence-mediating pathways. To add to the complexity, not all senescence-associated metabolic alterations are SnC specific. Cells such as neurons, cardiomyocytes, adipocytes and others, who are terminally differentiated and post mitotic, also exhibit senescence-associated phenotypes such as increased expression of p16^{INK4a} or other CDKIs, which may or may not reflect a truly senescent cell state. Also, the specific targeting of shared signalling pathways between senescent and non-senescent cells can result in off-target effects, creating barriers to clinical application. For example, navitoclax, a BCL-2 family inhibitor, is not only senolytic on certain cell types, but also exhibits severe cytotoxic effects towards platelets and neutrophils resulting in thrombocytopenia and neutropenia.

One key question is whether the induction of senolysis of all types of senescent cells is truly the best therapeutic approach. Apart from the possibility of off-target effects through senolytic targeting of shared critical pathways between SnCs and proliferating or quiescent cells, SnCs themselves have been found to play important roles in tissue renewal, wound healing, cancer prevention and embryonic development. Thus, it is critical that senotherapeutic treatment still permits beneficial senescent cell populations to persist.

However, since SnCs accumulate slowly overtime, senolytics are anticipated to be administered periodically, which may prevent dysregulation of critical senescent cell-driven processes. Even so, senomorphics, which attenuate their deleterious pathological, pro-inflammatory secretory phenotype without inducing senolysis in certain cases, may be a better therapeutic option. Senomorphics are envisioned to be administered daily to suppress the slowly growing population of SnCs. Since senomorphics still target senescence-associated pathways, we must ensure that these molecules do not have adverse effects. Attenuating certain senescence phenotypes for prolonged durations may invoke global cell cycle dysregulation and may prevent senescent cell regeneration, promoting cancer and dysregulation of important cellular processes previously mentioned. However, senomorphics, like senolytics, have been demonstrated to extend both healthspan and lifespan in animal models. They have also been shown to have side effects that result from off-target effects. Treatment with the senomorphic rapamycin, which has been shown to alleviate age-associated dysfunctions in model organisms, similar to navitoclax, also induces thrombocytopenia, metabolic dysregulation, hyperlipidaemia and impairs wound healing.

It is also important to note that many currently reported senotherapeutic molecules have only been reported to be examined at a singular concentration. Also, many cell culture experiments with senotherapeutics rely on measuring markers of senescence such as SASP and cyclin-dependent kinase inhibitors p16^{INK4a} and p21^{CIP1} as well as SA- β -gal activity. However, some of these senescence-associated traits may be shared with the proliferating or quiescent states of the cell and may also be differentially affected by senotherapeutic molecules, resulting in false positives and incorrect identification of mechanistic action. Therefore, future studies examining mechanistic action and potency of senotherapies need to first characterize the senescent cell types used in the assay and then utilize multiple markers associated with their respective senescent state.

The issues with defining senotherapeutics in cell culture are compounded in model organisms where the reduction of multiple markers of senescence is not sufficient to determine if a compound is senolytic or senomorphic. However, the effective treatment regimen is at least one indicator of mechanistic action where intermittent treatment resulting in prolonged reduction of senescence and SASP markers and increased health benefit is consistent with senolytic activity. Conversely, a compound requiring frequent treatment would denote senomorphic activity. These assays also harbour additional faults since they also neglect to

determine senotherapeutic specificity towards specific senescent cell types or sub-populations, and do not address whether the senotherapeutic benefits result from selectively targeting SnCs, or an indirect mechanism that improves the immune system's ability to clear senescent cells.

In summary, the development of senotherapeutics to treat age-related diseases and extend the period of healthy longevity is progressing at a rapid rate. The first senolytics were reported in 2015; now, there are more than 20 senolytic trials ongoing. Additionally, drugs known to have beneficial effects on healthspan such as metformin and rapamycin have been shown to have senotherapeutic activity. Thus, it is anticipated that this field of senotherapeutics will continue to expand with more optimal drugs and drug combinations identified, with many being tested in clinical trials.

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Conflict of interest

Drs Niedernhofer and Robbins are co-founders of NRTK Bioscience and have filed patents on senolytics and senomorphic compounds.

Author contributions

LZ, LEP and VP wrote the manuscript that was then edited by LJN and PDR.

Data availability statement

Data sharing is not applicable to this article as no new data were created in this study.

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